

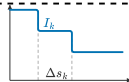
LLM-informed Warm Start

Battery-domain Knowledge

battery context



MCC
protocol
structure



$$\theta = [I_1, \dots, I_K, \Delta s_1, \dots, \Delta s_{K-1}]^T$$

$$f(\theta) = [t_c(\theta), T_p(\theta), Q_s(\theta)]^T$$

Operating
practice

$$I_1 \geq I_2 \geq I_3$$

$$\Delta s_1 + \Delta s_2 \leq 0.65$$

LLM reasoning

Propose 15 diverse 3-stage protocols
for 0–80% SOC

"high current at low SOC is safe;
taper toward full charge. spread
candidates across trade-
offs: fast/cool/long-life"

Initial Protocols

LLM output: $\theta_1, \theta_2, \dots, \theta_{15}$

candidate screening &
selection

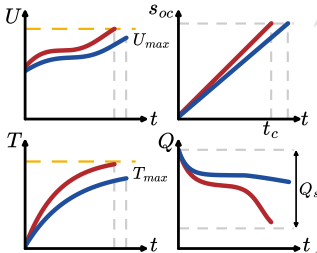
$$D_0 = \{(\theta_j, f(\theta_j))\}_{j=1}^{n_0}, n_0 = 6$$

$$P_0 = \text{feasible Pareto set of } D_0$$

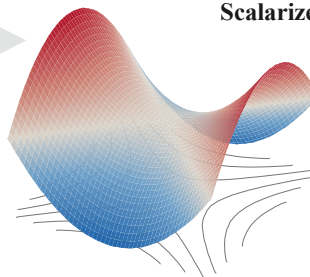
LLMBO-MO Loop

Battery Simulation

Thermal
Time
Degradation



Scalarized GP

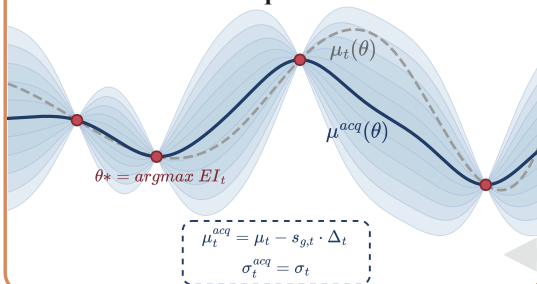


$$g_t^{aug}(\theta) = \rho_{aug} \sum_{i=1}^3 w_i^{(t)} \bar{f}_{i,t}(\theta)$$

$$g_t(\theta) = \max_{i=1}^3 \{w_i^{(t)} \bar{f}_{i,t}(\theta)\} + g_t^{aug}(\theta)$$

$$g(\theta) \sim \mathcal{N}(0, k(\theta, \theta')) \rightarrow \mu_t, \sigma_t$$

EI Acquisition



$$\mu_t^{acq} = \mu_t - s_{g,t} \cdot \Delta_t$$

$$\sigma_t^{acq} = \sigma_t$$

LLM Regional Guidance

"high early current saves time;
keep Δs_1 moderate
to limit plating risk. Propose
 $R_t : I_1 \in [4.1, 5.1] A$,
 $\Delta s_1 \in [0.24, 0.33], \hat{c}_t = 0.7$ "

$$\{R_t, \hat{c}_t\} = LLM(w^{(t)}, D_t), t \leq T_L$$

LLM preference density

